

Remux Mode (MKV → MP4)

Version: v3.0 · Status: Stable

What it does

Selecting **Remux only (Copy → MP4)** in the *Video Codec* dropdown rewraps an `.mkv` into an `.mp4` **without re-encoding** — same video bytes, same audio bytes, new container. A 50 GB UHD file finishes in under a minute on NVMe. Crucially, the **Dolby Vision RPU layer is preserved**, which is why this mode exists: many TVs (LG OLED, certain Samsungs, Apple TV) only render DV from MP4, not MKV — even though the underlying HEVC bitstream is identical.

How to use it

1. Load a source via the **Input** field.
2. Pick **Remux only (Copy → MP4)** in the *Video Codec* dropdown. All irrelevant controls (bitrate slider, audio codec, subtitles, 4K→1080p) gray out automatically.
3. Optionally uncheck audio tracks you don't want.
4. Press **Run**. Output lands next to the source as `.mp4`.

What gets transferred

Stream	Behavior
Video	Copied 1:1 — bitstream, DV RPU, HDR10/10+ metadata, timestamps
Audio	Copied 1:1 per checked track (codec must fit MP4)
Subtitles	Text subs (SRT/ASS/SSA) auto-converted to <code>mov_text</code> . Picture subs (PGS/DVB) cannot be muxed and are disabled in the list

Audio codec compatibility

Codec	Fits in MP4?	Notes
AAC, AC-3, E-AC-3, MP3	✓	Standard MP4 audio
FLAC, Opus	⚠	Spec-allowed; older hardware may choke
TrueHD	✗	Common in UHD-BluRay rips — not supported
DTS-HD MA	✗	Not supported in MP4

If the source has a TrueHD track, the GUI shows a warning before launch. Deselect TrueHD tracks and keep only AC-3 / E-AC-3, or fall back to a re-encode mode if TrueHD is the only audio.

Audio-only extraction

Unchecking **Include Video** turns the dialog into an audio extractor. The output extension switches automatically to `.mka` (Copy) or the format-specific extension when the **Audio output format** dropdown is changed:

Format	Extension	Behavior
Copy	<code>.mka</code>	Container-only — original codec preserved, no quality change
FLAC	<code>.flac</code>	Lossless 24-bit re-encode (<code>-sample_fmt s32 -bits_per_raw_sample 24</code>)
MP3	<code>.mp3</code>	Lossy, libmp3lame at FFmpeg defaults
AAC	<code>.m4a</code>	Lossy, libfdk_aac / native AAC at FFmpeg defaults

FFmpeg's filter chain runs in 32-bit float internally — there's no resolution loss from the decode/encode round-trip beyond what the source codec itself already lost. (AC-3/E-AC-3 are lossy formats; FLAC stores the decoded samples faithfully but cannot recover what the lossy encoder discarded.)

Multi-channel downmix

The **Downmix audio to:** checkbox folds surround tracks into a smaller layout. Available targets: `2.0 (Stereo)` and `5.1`. The `5.1` option is only selectable when at least one source track has more than 6 channels.

Music vs Movie mode

The mix coefficients differ depending on source material — choose the mode that matches what you're converting. **Default is Music** (the more common case for stereo downmixing) and the choice is remembered across sessions.

Music mode — full ITU-R BS.775 weighted mix:

```
L_out = FL + 0.707·FC + 0.707·SL + 0.707·LFE → volume -3 dB
R_out = FR + 0.707·FC + 0.707·SR + 0.707·LFE → volume -3 dB
```

Preserves bass energy and dynamics. Best for multi-channel music releases (Blu-ray Audio, SACD rips, surround music) where the LFE is typically a discrete bass channel, not an effect channel. The trailing `volume=-3dB` guards against clipping when all channels carry correlated signal.

Movie mode — conservative cinema downmix:

```
L_out = FL + 0.707·FC + 0.5·SL + 0.316·LFE
R_out = FR + 0.707·FC + 0.5·SR + 0.316·LFE
```

Center stays at -3 dB for dialogue clarity, surrounds pulled back to -6 dB so effects don't overwhelm the front mix, LFE attenuated to -10 dB to keep action-bass under control. Standard cinema downmix levels.

Per-track logic

Only tracks whose source channels exceed the target are re-encoded. A stereo track in the same selection stays as Copy. Mixed selections work without manual intervention.

Codec & bitrate

- *Video included*: source codec is preserved (AC-3 5.1 → AC-3 2.0, E-AC-3 5.1 → E-AC-3 2.0).
- *Audio-only*: the **Audio output format** dropdown decides — Copy keeps the source codec; FLAC/MP3/AAC override it.
- `Bitrate = min(codec_max, source_bitrate)` — uses the codec's maximum but never exceeds the source. AC-3 caps at 640 kbps, E-AC-3 at 1536 kbps, AAC at 512 kbps, MP3 at 320 kbps. FLAC has no bitrate setting (lossless).

Tip: avoid double-lossy on music

When **Music mode** is active and the source is lossy (AC-3 / E-AC-3 / AAC), the GUI shows a hint: "*switch Audio output format to FLAC for lossless music quality.*" This prevents a second lossy encode. The original lossy artifacts remain (FLAC can't recover what AC-3 discarded), but no further quality is lost in the downmix step itself.

The FFmpeg command

Standard remix (video included, all audio copy):

```
ffmpeg -y -progress pipe:1 -loglevel error
-i "source.mkv"
-map 0:v:0 -map 0:1 -map 0:2
-c:v copy -c:a copy -strict -2
"source.mp4"
```

Audio-only with Music-mode 5.1 → 2.0 downmix to FLAC:

```
ffmpeg -y -i "source.mkv"
-map 0:1
-c:a:0 flac
-filter:a:0 "pan=stereo|FL=c0+0.707*c2+0.707*c4+0.707*c3|FR=c1+0.707*c2+0.707*c5+0.
-sample_fmt:a:0 s32 -bits_per_raw_sample 24
"source.flac"
```

The `c0..c5` channel notation in `pan` is robust against 5.1 vs 5.1(side) layout differences — FL/FR/FC/LFE are constant, but BL/SL naming can vary. `-strict -2` is defensive and explicitly allows E-AC-3 in MP4. No `-c:s` is emitted unless subtitle streams are mapped (then `-c:s mov_text`).

When NOT to remix

- **Reducing file size** — remix preserves bitrate exactly. Re-encode instead.
- **Changing resolution** — remix can't downscale. Use H.265/AV1 mode.
- **Source has TrueHD-only audio or MPEG-2 video** — won't fit in MP4.
- **Your player already handles DV in MKV** — modern Apple TV, recent firmware on many smart TVs do. Test playback first.

Troubleshooting

- **Output a few hundred KB different in size**: normal — container framing overhead differs between MKV and MP4. The streams are byte-identical.
- **MP4 plays but not as DV**: likely a DV profile mismatch (profile 7 dual-layer is the most fragile; profiles 5 and 8.1 are widely supported), or the source was HDR10 only — verify with MediaInfo.
- **"codec not currently supported in container"**: TrueHD or MPEG-2 in the source. Deselect the

offending track or re-encode.

- **Progress finishes in 5 seconds:** expected. The file is on disk — check next to the input.
- **Downmix checkbox stays disabled:** no track in the current selection has more than 2 channels. Hover for the tooltip explanation.
- **Stereo downmix sounds thin or quiet:** make sure Music mode is selected for music sources. FFmpeg's default `-ac 2` (which we *don't* use) is much more attenuated than the explicit ITU-R mix this tool now applies.